

Comprehensive Practice: 60 Questions

PHY 104 Oscillations, Normal Modes, Waves and Optics

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Instructions: Choose the best answer in each case. Questions 1—20 test foundations, Questions 21—40 test applied reasoning, and Questions 41—60 are multi-step challenges. Work independently before consulting the answer key and fully worked solutions.

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1 Question Booklet

Where required, take: $g = 9.80 \text{ m s}^{-2}$, $v_{\text{sound}} = 343 \text{ m s}^{-1}$ unless otherwise stated, $c = 3.00 \times 10^8 \text{ m s}^{-1}$, $\rho_{\text{air}} = 1.21 \text{ kg m}^{-3}$, $I_0 = 1.00 \times 10^{-12} \text{ W m}^{-2}$. Angles are measured from the normal unless stated otherwise.

Level I: Foundational Reasoning — Questions 1—20

- A particle is displaced a distance $x > 0$ from equilibrium and executes simple harmonic motion. Which expression correctly gives the restoring force and its direction?
 - $F = +kx$, directed away from equilibrium
 - $F = -kx$, directed away from equilibrium
 - $F = -kx$, directed toward equilibrium
 - $F = -k/x$, directed toward equilibrium
 - $F = 0$ because the particle is momentarily displaced
- A 0.50 kg mass is attached to a spring of constant 200 N m^{-1} . What is the period of the motion?
 - 0.157 s
 - 0.314 s
 - 0.628 s
 - 1.57 s
 - 3.14 s
- A platform oscillates with amplitude 4.0 cm and frequency 2.5 Hz . What is its maximum speed?
 - 0.100 m s^{-1}
 - 0.251 m s^{-1}
 - 0.400 m s^{-1}
 - 0.628 m s^{-1}
 - 1.57 m s^{-1}
- A mass—spring oscillator has amplitude A and total mechanical energy E . When the displacement is $x = A/2$, which statement is correct?
 - $U = E/2$ and $K = E/2$
 - $U = E/4$ and $K = 3E/4$
 - $U = 3E/4$ and $K = E/4$
 - $U = E$ and $K = 0$
 - $U = 0$ and $K = E$
- What length must a simple pendulum have to possess a period of 2.00 s near Earth's surface?
 - 0.993 m
 - 1.56 m
 - 2.00 m
 - 3.12 m
 - 9.80 m
- Which statement best defines a normal mode of a coupled system?
 - Every part remains at rest while energy moves through the system.
 - Every part oscillates with one angular frequency and fixed phase relations.
 - Different parts oscillate at unrelated frequencies.
 - Only one part of the system is allowed to move.
 - The coupling force must vanish at every instant.
- Two identical masses m are attached to end springs of constant k and joined by a coupling spring k_c . What is the angular frequency of the in-phase mode?
 - $\sqrt{k/m}$
 - $\sqrt{k_c/m}$
 - $\sqrt{(k + k_c)/m}$
 - $\sqrt{(k + 2k_c)/m}$
 - $\sqrt{2k/m}$
- Which amplitude vector represents the out-of-phase normal mode of two identical coupled oscillators?
 - $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$
 - $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$
 - $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
 - $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
 - $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$
- Why is the out-of-phase normal frequency higher than the in-phase normal frequency in the symmetric two-mass system?

- A. The masses become heavier in the out-of-phase mode.
- B. The end springs stop exerting forces.
- C. The coupling spring is strongly stretched or compressed and adds restoring force.
- D. The coupling spring removes kinetic energy permanently.
- E. The amplitude vector is not an eigenvector.
10. For $k = 100 \text{ N m}^{-1}$ and $k_c = 50 \text{ N m}^{-1}$, what is the ratio ω_2/ω_1 ?
- A. 1.00
- B. $\sqrt{2} \approx 1.41$
- C. 2.00
- D. $\sqrt{3} \approx 1.73$
- E. 0.707
11. A transverse wave is described by $y = (3.0 \text{ mm}) \sin(4x - 20t)$, with SI units. Which pair gives its wavelength and speed?
- A. $\lambda = 0.250 \text{ m}$, $v = 80 \text{ m s}^{-1}$
- B. $\lambda = 1.00 \text{ m}$, $v = 5.0 \text{ m s}^{-1}$
- C. $\lambda = 1.57 \text{ m}$, $v = 5.0 \text{ m s}^{-1}$
- D. $\lambda = 1.57 \text{ m}$, $v = 80 \text{ m s}^{-1}$
- E. $\lambda = 6.28 \text{ m}$, $v = 0.20 \text{ m s}^{-1}$
12. The wave $y = A \cos(6x + 24t)$ travels in which direction and at what speed?
- A. $+x$ at 24 m s^{-1}
- B. $+x$ at 4 m s^{-1}
- C. $-x$ at 24 m s^{-1}
- D. $-x$ at 4 m s^{-1}
- E. It is a standing wave.
13. A standing wave has wavelength 0.80 m . What is the separation between adjacent nodes?
- A. 0.20 m
- B. 0.40 m
- C. 0.80 m
- D. 1.20 m
- E. 1.60 m
14. A string of length 0.80 m , fixed at both ends, carries waves at 120 m s^{-1} . What is the third-harmonic frequency?
- A. 75 Hz
- B. 120 Hz
- C. 150 Hz
- D. 180 Hz
- E. 225 Hz
15. A sound intensity is $1.0 \times 10^{-6} \text{ W m}^{-2}$. What is the corresponding sound level? Take $I_0 = 1.0 \times 10^{-12} \text{ W m}^{-2}$.
- A. 6 dB
- B. 12 dB
- C. 60 dB
- D. 100 dB
- E. 120 dB
16. Light of vacuum wavelength 600 nm enters a medium of refractive index 1.50 . Which pair gives its speed and wavelength in the medium?
- A. $2.00 \times 10^8 \text{ m s}^{-1}$ and 400 nm
- B. $2.00 \times 10^8 \text{ m s}^{-1}$ and 600 nm
- C. $3.00 \times 10^8 \text{ m s}^{-1}$ and 400 nm
- D. $4.50 \times 10^8 \text{ m s}^{-1}$ and 900 nm
- E. $1.50 \times 10^8 \text{ m s}^{-1}$ and 300 nm
17. A ray passes from air into glass of refractive index 1.50 at an incidence angle of 30.0° . What is the refracted angle?
- A. 10.0°
- B. 15.0°
- C. 30.0°
- D. 19.5°
- E. 48.6°
18. In a Young double-slit experiment, $\lambda = 600 \text{ nm}$, the screen distance is 2.0 m , and the slit separation is 0.40 mm . What is the fringe spacing?
- A. 0.75 mm
- B. 3.00 mm
- C. 6.00 mm
- D. 12.0 mm
- E. 30.0 mm
19. Light of wavelength 500 nm passes through a single slit of width 0.20 mm . At what angle does the first minimum occur?

- A. 0.0143°
 B. 0.0286°
 C. 0.0716°
 D. 0.100°
 E. 0.143°
20. A telescope of aperture diameter 0.10 m observes light of wavelength 550 nm. What is its diffraction-limited angular resolution?
- A. 5.50×10^{-7} rad
 B. 4.51×10^{-6} rad
 C. 6.71×10^{-6} rad
 D. 1.22×10^{-5} rad
 E. 6.71×10^{-5} rad

Level II: Applied Problems — Questions
21—40

21. An oscillator obeys $x = A \cos(\omega t + \phi)$. At $t = 0$, $x = 3.0$ cm and $v = -0.40$ m s $^{-1}$, while $\omega = 10$ rad s $^{-1}$. What are A and ϕ ?
- A. $A = 4.0$ cm, $\phi = 36.9^\circ$
 B. $A = 5.0$ cm, $\phi = 53.1^\circ$
 C. $A = 5.0$ cm, $\phi = -53.1^\circ$
 D. $A = 7.0$ cm, $\phi = 45.0^\circ$
 E. $A = 3.0$ cm, $\phi = 90.0^\circ$
22. A uniform rod of length 1.20 m swings with small amplitude about a horizontal pivot through one end. What is its period?
- A. 0.898 s
 B. 1.27 s
 C. 1.55 s
 D. 1.80 s
 E. 2.20 s
23. Three identical springs, each of constant 120 N m $^{-1}$, are connected in series to a 0.36 kg mass. What is the oscillation period?
- A. 0.172 s
 B. 0.344 s
 C. 0.596 s
 D. 1.03 s
 E. 1.88 s

24. A 10.0 g bullet moving at 400 m s $^{-1}$ embeds in a 1.99 kg block attached to a spring of constant 800 N m $^{-1}$. The block is initially at equilibrium and at rest. Which pair gives the speed immediately after impact and the resulting amplitude?

- A. 2.00 m s $^{-1}$ and 0.100 m
 B. 4.00 m s $^{-1}$ and 0.200 m
 C. 2.01 m s $^{-1}$ and 0.050 m
 D. 1.00 m s $^{-1}$ and 0.100 m
 E. 200 m s $^{-1}$ and 0.500 m

25. A torsional oscillator has torsion constant

$$\kappa = 0.80 \text{ N m rad}^{-1},$$

moment of inertia $I = 0.020$ kg m 2 , and angular amplitude $\theta_{\max} = 0.25$ rad. What is its angular speed when $\theta = 0.15$ rad?

- A. 0.400 rad s $^{-1}$
 B. 0.800 rad s $^{-1}$
 C. 1.00 rad s $^{-1}$
 D. 1.60 rad s $^{-1}$
 E. 1.26 rad s $^{-1}$

26. For two identical coupled oscillators, $m = 0.50$ kg, $k = 100$ N m $^{-1}$, and $k_c = 50$ N m $^{-1}$. What are the two normal angular frequencies?

- A. 10.0 and 14.1 rad s $^{-1}$
 B. 14.1 and 20.0 rad s $^{-1}$
 C. 20.0 and 28.3 rad s $^{-1}$
 D. 7.07 and 14.1 rad s $^{-1}$
 E. 14.1 and 17.3 rad s $^{-1}$

27. A symmetric two-mass system has $m = 2.0$ kg and measured normal angular frequencies $\omega_1 = 10$ rad s $^{-1}$ and $\omega_2 = 14$ rad s $^{-1}$. Find k and k_c .

- A. $k = 100$ N m $^{-1}$, $k_c = 48$ N m $^{-1}$
 B. $k = 196$ N m $^{-1}$, $k_c = 100$ N m $^{-1}$
 C. $k = 200$ N m $^{-1}$, $k_c = 192$ N m $^{-1}$
 D. $k = 200$ N m $^{-1}$, $k_c = 96$ N m $^{-1}$
 E. $k = 392$ N m $^{-1}$, $k_c = 96$ N m $^{-1}$

28. The initial displacement of two identical coupled masses is $\mathbf{x}(0) = \begin{bmatrix} A \\ 0 \end{bmatrix}$, with zero initial velocities. In the expansion $\mathbf{x} = C_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + C_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, what are C_1 and C_2 at $t = 0$?

- A. $C_1 = A$, $C_2 = 0$
- B. $C_1 = 0$, $C_2 = A$
- C. $C_1 = A/2$, $C_2 = A/2$
- D. $C_1 = A$, $C_2 = -A$
- E. $C_1 = 2A$, $C_2 = 2A$
29. For $x_1 = 3.0\text{ cm}$ and $x_2 = -1.0\text{ cm}$, what are the normal coordinates $q_1 = x_1 + x_2$ and $q_2 = x_1 - x_2$?
- A. $q_1 = 2.0\text{ cm}$, $q_2 = 4.0\text{ cm}$
- B. $q_1 = 4.0\text{ cm}$, $q_2 = 2.0\text{ cm}$
- C. $q_1 = 2.0\text{ cm}$, $q_2 = -4.0\text{ cm}$
- D. $q_1 = -2.0\text{ cm}$, $q_2 = 4.0\text{ cm}$
- E. $q_1 = 3.0\text{ cm}$, $q_2 = -1.0\text{ cm}$
30. Which characteristic equation determines the normal angular frequencies of the symmetric two-mass system?
- A. $k + k_c - m\omega^2 = 0$
- B. $(k - m\omega^2)^2 + k_c^2 = 0$
- C. $(k + k_c + m\omega^2)^2 - k_c^2 = 0$
- D. $(k + 2k_c - m\omega^2)^2 = 0$
- E. $(k + k_c - m\omega^2)^2 - k_c^2 = 0$
31. A string wave is $y = (4.0\text{ mm})\sin(5x - 100t)$. If the linear density is 0.010 kg m^{-1} , which set gives the wave speed, tension, and average power?
- A. 5 m s^{-1} , 0.25 N , 0.004 W
- B. 20 m s^{-1} , 4.0 N , 0.016 W
- C. 20 m s^{-1} , 2.0 N , 0.032 W
- D. 100 m s^{-1} , 100 N , 0.080 W
- E. 500 m s^{-1} , 2500 N , 1.60 W
32. A pipe of length 0.85 m is closed at one end and open at the other. Taking the sound speed as 340 m s^{-1} , what are its first three resonant frequencies?
- A. 100, 200, 300 Hz
- B. 200, 400, 600 Hz
- C. 100, 300, 500 Hz
- D. 50, 150, 250 Hz
- E. 100, 400, 700 Hz
33. A stretched wire has fundamental frequency 500 Hz . Its tension is increased slightly so that it produces 5.0 beats s^{-1} with the original wire. What fractional increase in tension is required?
- A. 0.0050
- B. 0.0100
- C. 0.0150
- D. 0.0201
- E. 0.0404
34. A source emitting 600 Hz moves directly toward a stationary observer at 30 m s^{-1} . Taking $v = 343\text{ m s}^{-1}$, what frequency is observed?
- A. 658 Hz
- B. 600 Hz
- C. 552 Hz
- D. 630 Hz
- E. 714 Hz
35. An aircraft travels at Mach 1.50. What is the Mach-cone half-angle?
- A. 30.0°
- B. 33.7°
- C. 45.0°
- D. 60.0°
- E. 41.8°
36. A mica sheet of refractive index 1.60 and thickness $5.0\text{ }\mu\text{m}$ is placed over one slit in a double-slit experiment using 600 nm light. By how many fringe spacings does the pattern shift?
- A. 3
- B. 5
- C. 6
- D. 8
- E. 10
37. In a real double-slit arrangement, the slit separation is four times the slit width: $d = 4a$. Which interference orders are missing?
- A. $m = 1, 2, 3, \dots$
- B. $m = 2, 4, 6, \dots$
- C. $m = 4, 8, 12, \dots$
- D. $m = 3, 6, 9, \dots$
- E. No orders are missing.
38. A diffraction grating has 500 lines per millimetre and is illuminated with 600 nm light. Which pair gives the second-order angle and the maximum possible order?

- A. 17.5° and $m_{\max} = 2$
 B. 30.0° and $m_{\max} = 3$
 C. 53.1° and $m_{\max} = 4$
 D. 36.9° and $m_{\max} = 3$
 E. 90.0° and $m_{\max} = 2$
39. For glass of refractive index 1.50 surrounded by air, which pair gives (i) the critical angle for glass-to-air incidence and (ii) Brewster's angle for air-to-glass incidence?
 A. 41.8° and 56.3°
 B. 56.3° and 41.8°
 C. 33.7° and 48.2°
 D. 48.2° and 33.7°
 E. 90.0° and 45.0°
40. X-rays of wavelength 0.154 nm produce a first-order Bragg peak at a measured scattering angle $2\theta = 50.0^\circ$. What is the crystal-plane spacing?
 A. 0.0770 nm
 B. 0.119 nm
 C. 0.154 nm
 D. 0.238 nm
 E. 0.182 nm
- Level III: Multi-step Challenges — Questions**
41—60
41. An oscillator has amplitude 6.0 cm and angular frequency 8.0 rad s^{-1} . At $t = 0$, its displacement is 3.0 cm and it is moving in the positive direction. Which equation is correct?
 A. $x = 0.060 \cos(8t + \pi/3)$
 B. $x = 0.030 \cos(8t - \pi/3)$
 C. $x = 0.060 \cos(8t - \pi/3)$
 D. $x = 0.060 \sin(8t - \pi/3)$
 E. $x = 0.060 \cos(8t + 2\pi/3)$
42. At a displacement of 3.0 cm, a mass in SHM has speed 0.40 m s^{-1} . Its turning points are at $x = \pm 5.0 \text{ cm}$. What is the period?
 A. 0.314 s
 B. 0.628 s
 C. 0.785 s
 D. 1.26 s
 E. 1.57 s
43. A uniform rod of length 1.20 m oscillates as a physical pendulum about a pivot located one-quarter of its length from one end. What is its small-angle period?
 A. 0.840 s
 B. 1.12 s
 C. 1.44 s
 D. 1.68 s
 E. 2.06 s
44. At $t = 0$, an oscillator is at $x = A/2$ and moving toward equilibrium. How long does it first take to reach the opposite turning point $x = -A$?
 A. $T/3$
 B. $T/4$
 C. $T/2$
 D. $2T/3$
 E. $3T/4$
45. A 1.00 kg block attached to a 100 N m^{-1} spring passes through equilibrium at 2.00 m s^{-1} . At that instant, a 0.25 kg lump of clay moving vertically falls onto the block and sticks. What is the new horizontal oscillation amplitude?
 A. 0.100 m
 B. 0.125 m
 C. 0.160 m
 D. 0.200 m
 E. 0.179 m
46. For a symmetric coupled system with $m = 1 \text{ kg}$, $k = 4 \text{ N m}^{-1}$, and $k_c = 6 \text{ N m}^{-1}$, mass 1 is initially displaced by A , mass 2 is at equilibrium, and both are released from rest. What are the displacements at $t = \pi/2 \text{ s}$?
 A. $x_1 = A, x_2 = 0$
 B. $x_1 = -A, x_2 = 0$
 C. $x_1 = 0, x_2 = -A$
 D. $x_1 = 0, x_2 = A$
 E. $x_1 = A/2, x_2 = -A/2$
47. Two identical masses of 0.50 kg have normal frequencies $f_1 = 2.00 \text{ Hz}$ and $f_2 = 3.00 \text{ Hz}$. What are k and k_c ?

- A. $k = 19.7 \text{ N m}^{-1}$, $k_c = 12.3 \text{ N m}^{-1}$
 B. $k = 79.0 \text{ N m}^{-1}$, $k_c = 49.3 \text{ N m}^{-1}$
 C. $k = 78.9 \text{ N m}^{-1}$, $k_c = 98.7 \text{ N m}^{-1}$
 D. $k = 157.9 \text{ N m}^{-1}$, $k_c = 49.3 \text{ N m}^{-1}$
 E. $k = 39.5 \text{ N m}^{-1}$, $k_c = 24.7 \text{ N m}^{-1}$
48. One mass of a weakly coupled identical pair is displaced and both are released from rest. If the normal angular frequencies are 10 rad s^{-1} and 12 rad s^{-1} , when does the first complete transfer of oscillation amplitude to the other mass occur?
 A. 0.262 s
 B. 0.524 s
 C. 1.05 s
 D. 1.57 s
 E. 3.14 s
49. For the same displacement amplitude A of each mass, compare the total energies of the pure out-of-phase and in-phase modes when $k = 100 \text{ N m}^{-1}$ and $k_c = 50 \text{ N m}^{-1}$. What is $E_{\text{out}}/E_{\text{in}}$?
 A. 2.00
 B. 1.50
 C. 1.00
 D. 0.50
 E. 3.00
50. A system with unit masses has stiffness matrix $K = \begin{bmatrix} 6 & -2 \\ -2 & 6 \end{bmatrix} \text{ N.m}^{-1}$. Its initial displacement is $\begin{bmatrix} 3 \\ 1 \end{bmatrix} \text{ cm}$. Which option gives the modal coefficients in the basis $[1, 1]^T, [1, -1]^T$ and the corresponding angular frequencies?
 A. $C_1 = 1 \text{ cm}$, $C_2 = 2 \text{ cm}$; $\omega_1 = \sqrt{6}$, $\omega_2 = \sqrt{2}$
 B. $C_1 = 3 \text{ cm}$, $C_2 = 1 \text{ cm}$; $\omega_1 = 2$, $\omega_2 = 4$
 C. $C_1 = 2 \text{ cm}$, $C_2 = -1 \text{ cm}$; $\omega_1 = \sqrt{8}$, $\omega_2 = 2$
 D. $C_1 = 4 \text{ cm}$, $C_2 = 2 \text{ cm}$; $\omega_1 = 4$, $\omega_2 = 8$
 E. $C_1 = 2 \text{ cm}$, $C_2 = 1 \text{ cm}$; $\omega_1 = 2$, $\omega_2 = 2\sqrt{2}$
51. Three waves of equal frequency and amplitude 5.0 mm have phase constants 0, $\pi/3$, and $2\pi/3$. What are the amplitude and phase constant of their resultant?
 A. 5.0 mm and 0
 B. 8.66 mm and $\pi/2$
 C. 10.0 mm and $\pi/3$
 D. 15.0 mm and $\pi/3$
 E. 0 and an undefined phase
52. A stone is dropped from rest into a well. The splash is heard 4.00 s later. Taking $g = 9.80 \text{ m s}^{-2}$ and the sound speed as 343 m s^{-1} , what is the well depth?
 A. 61.0 m
 B. 70.5 m
 C. 78.4 m
 D. 86.0 m
 E. 39.2 m
53. A stationary detector emits ultrasound at 40.0 kHz toward a car approaching at 30.0 m s^{-1} . The wave reflects from the car and returns to the detector. Taking $v = 343 \text{ m s}^{-1}$, what return frequency is detected?
 A. 36.8 kHz
 B. 40.0 kHz
 C. 43.5 kHz
 D. 45.2 kHz
 E. 47.7 kHz
54. A standing wave is $y = 0.080 \sin(\pi x/2) \cos(4\pi t)$ in SI units. At $x = 0.50 \text{ m}$ and $t = 0.0625 \text{ s}$, which pair gives the local amplitude and instantaneous displacement?
 A. 56.6 mm and 40.0 mm
 B. 80.0 mm and 56.6 mm
 C. 40.0 mm and 56.6 mm
 D. 56.6 mm and 0
 E. 40.0 mm and 40.0 mm
55. Two coherent in-phase point sources are separated by 1.50 m and emit wavelength 0.400 m. A detector moves once around a very large circle in the plane of the sources. How many intensity maxima does it encounter?
 A. 7
 B. 8
 C. 12
 D. 14
 E. 16

56. For a real double slit, the intensity is

$$I = I_0(\sin \beta / \beta)^2 \cos^2 \alpha.$$

At an angle where $\beta = \pi/2$ and $\alpha = \pi/3$, what is I/I_0 ?

- A. 1/4
 B. $1/\pi^2 \approx 0.101$
 C. $4/\pi^2 \approx 0.405$
 D. 1/2
 E. 0
57. A grating must just resolve 600.0 nm and 600.3 nm in second order. If the grating has 500 lines per millimetre, what minimum number of illuminated lines and illuminated width are required?
- A. $N = 500$ and 1.00 mm
 B. $N = 667$ and 1.33 mm
 C. $N = 1000$ and 2.00 mm
 D. $N = 2000$ and 4.00 mm
 E. $N = 4000$ and 8.00 mm
58. For a single slit of width 0.20 mm illuminated by 600 nm light, the first secondary maximum occurs at $\beta \approx 4.493$. Approximately where is it, and what fraction of the central intensity does it have?
- A. 0.086° and 50%
 B. 0.143° and 10%
 C. 0.200° and 1.0%
 D. 0.246° and 4.7%
 E. 0.492° and 0.47%
59. Two headlights separated by 1.50 m are 100 km away. What minimum telescope aperture is required to just resolve them at 550 nm, according to Rayleigh's criterion?
- A. 4.47 cm
 B. 2.24 cm
 C. 8.94 cm
 D. 44.7 cm
 E. 1.22 cm
60. A grating has 300 lines per millimetre, and each slit has width $a = d/3$, where d is the grating spacing. It is illuminated with 500 nm light. How many observable principal maxima occur over both sides, including the central maximum, after missing orders are excluded?

- A. 5
 B. 7
 C. 11
 D. 13
 E. 9

2 Answer Key and Topic Map

Answer Key

1. C	2. B	3. D	4. B	5. A
6. B	7. A	8. D	9. C	10. B
11. C	12. D	13. B	14. E	15. C
16. A	17. D	18. B	19. E	20. C
21. B	22. D	23. C	24. A	25. E
26. B	27. D	28. C	29. A	30. E
31. B	32. C	33. D	34. A	35. E
36. B	37. C	38. D	39. A	40. E
41. C	42. B	43. D	44. A	45. E
46. C	47. B	48. D	49. A	50. E
51. C	52. B	53. E	54. A	55. D
56. B	57. C	58. D	59. A	60. E

Topic distribution: Each topic contributes 15 questions: Oscillations (1–5, 21–25, 41–45), Normal Modes (6–10, 26–30, 46–50), Waves and Sound (11–15, 31–35, 51–55), and Optics (16–20, 36–40, 56–60).

3 Fully Worked Solutions

3.1 Level I: Foundational Reasoning

Question 1 — Correct option C

Topic: Oscillations.

For simple harmonic motion, the defining force law is Hooke's law,

$$F = -kx.$$

The magnitude kx is proportional to the displacement. The minus sign does not mean that the force is always numerically negative; it means that the force points opposite to the displacement. Since $x > 0$, the restoring force points in the negative direction, toward equilibrium. Hence option **C**.

Question 2 — Correct option B

Topic: Oscillations.

For a mass-spring oscillator,

$$T = 2\pi\sqrt{\frac{m}{k}}.$$

Substituting $m = 0.50 \text{ kg}$ and $k = 200 \text{ N m}^{-1}$,

$$T = 2\pi\sqrt{\frac{0.50}{200}} = 2\pi\sqrt{0.0025} = 2\pi(0.0500) = 0.314 \text{ s}.$$

Therefore, option **B** is correct.

Question 3 — Correct option D

Topic: Oscillations.

The maximum speed in SHM is

$$v_{\max} = \omega A = 2\pi f A.$$

Here, $A = 0.040 \text{ m}$ and $f = 2.5 \text{ Hz}$, so

$$v_{\max} = 2\pi(2.5)(0.040) = 0.628 \text{ m s}^{-1}.$$

Thus, option **D**.

Question 4 — Correct option B

Topic: Oscillations.

For a spring oscillator,

$$E = \frac{1}{2}kA^2, \quad U = \frac{1}{2}kx^2.$$

At $x = A/2$,

$$U = \frac{1}{2}k\left\{\frac{A}{2}\right\}^2 = \frac{1}{8}kA^2 = \frac{1}{4}\left(\frac{1}{2}kA^2\right) = \frac{E}{4}.$$

Energy conservation gives $K = E - U = 3E/4$. Hence option **B**.

Question 5 — Correct option A

Topic: Oscillations.

For a small-amplitude simple pendulum,

$$T = 2\pi\sqrt{\frac{L}{g}}.$$

Solving for L ,

$$L = g\left(\frac{T}{2\pi}\right)^2 = 9.80\left(\frac{2.00}{2\pi}\right)^2 = 0.993 \text{ m}.$$

Therefore, option **A**.

Question 6 — Correct option B**Topic: Normal Modes.**

A normal mode is a special coordinated motion. Every moving part has the same angular frequency, while the phase relation and amplitude ratio between parts remain fixed. The parts don't necessarily have equal amplitudes, and the coupling force need not always vanish. Hence option **B**.

Question 7 — Correct option A**Topic: Normal Modes.**

In the in-phase mode, $x_1 = x_2$. The coupling spring has no change in length because both masses move together (in the same direction). It therefore adds no restoring force. Each mass behaves as though attached only to its outer spring, giving

$$\omega_1 = \sqrt{\frac{k}{m}}.$$

Thus, option **A**.

Question 8 — Correct option D**Topic: Normal Modes.**

Out-of-phase motion requires equal and opposite displacements:

$$x_1 = -x_2.$$

The vector $[1, -1]^T$ encodes precisely this relation. Any nonzero scalar multiple, such as $[3, -3]^T$, describes the same mode shape.

I may write vectors like $[1, 1]^T$, instead of the usual $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Therefore, option **D**.

Question 9 — Correct option C**Topic: Normal Modes.**

When the masses move oppositely, their relative displacement is large. The middle spring is therefore stretched or compressed substantially, so it contributes an additional restoring force. A stronger effective restoring action produces a larger angular frequency. Hence option **C**.

Question 10 — Correct option B**Topic: Normal Modes.**

The two angular frequencies are

$$\omega_1 = \sqrt{\frac{k}{m}}, \quad \omega_2 = \sqrt{\frac{k + 2k_c}{m}}.$$

Thus,

$$\frac{\omega_2}{\omega_1} = \sqrt{\frac{k + 2k_c}{k}} = \sqrt{\frac{100 + 2(50)}{100}} = \sqrt{2}.$$

Therefore, option **B**.

Question 11 — Correct option C**Topic: Waves.**

Comparing with $y = A \sin(kx - \omega t)$ gives $k = 4 \text{ rad m}^{-1}$ and $\omega = 20 \text{ rad s}^{-1}$. Hence,

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{4} = 1.57 \text{ m}, \quad v = \frac{\omega}{k} = \frac{20}{4} = 5.0 \text{ m s}^{-1}.$$

Thus, option **C**.

Question 12 — Correct option D**Topic: Waves.**A travelling wave of the form $kx + \omega t$ moves in the negative x -direction. Its speed is

$$v = \frac{\omega}{k} = \frac{24}{6} = 4 \text{ m s}^{-1}.$$

Therefore, option **D**.**Question 13 — Correct option B****Topic: Waves.**

Adjacent nodes of a standing wave are separated by half a wavelength:

$$\Delta x_{\text{node-node}} = \frac{\lambda}{2} = \frac{0.80}{2} = 0.40 \text{ m}.$$

Hence option **B**.**Question 14 — Correct option E****Topic: Waves.**

For a string fixed at both ends,

$$f_n = \frac{nv}{2L}.$$

With $n = 3$,

$$f_3 = \frac{3(120)}{2(0.80)} = \frac{360}{1.60} = 225 \text{ Hz}.$$

Thus, option **E**.**Question 15 — Correct option C****Topic: Waves.**

The sound level is

$$\beta = 10 \log_{10} \left(\frac{I}{I_0} \right).$$

Here, $I/I_0 = 10^{-6}/10^{-12} = 10^6$, so

$$\beta = 10 \log_{10}(10^6) = 10(6) = 60 \text{ dB}.$$

Therefore, option **C**.**Question 16 — Correct option A****Topic: Optics.**

The speed is

$$v = \frac{c}{n} = \frac{3.00 \times 10^8}{1.50} = 2.00 \times 10^8 \text{ m s}^{-1}.$$

The frequency is fixed by the source and does not change at the boundary. Hence,

$$\lambda = \frac{\lambda_0}{n} = \frac{600}{1.50} = 400 \text{ nm}.$$

Thus, option **A**.**Question 17 — Correct option D****Topic: Optics.**

Snell's law gives

$$n_1 \sin \theta_1 = n_2 \sin \theta_2.$$

With $n_1 = 1.00$, $n_2 = 1.50$, and $\theta_1 = 30.0^\circ$,

$$\sin \theta_2 = \frac{\sin 30^\circ}{1.50} = \frac{0.5}{1.5} = \frac{1}{3}.$$

Therefore, $\theta_2 = \sin^{-1}(1/3) = 19.5^\circ$, so option **D**.

Question 18 — Correct option B**Topic: Optics.**

For small angles, the double-slit fringe spacing is

$$\Delta y = \frac{\lambda L}{d}.$$

Thus,

$$\Delta y = \frac{(600 \times 10^{-9})(2.0)}{0.40 \times 10^{-3}} = 3.00 \times 10^{-3} \text{ m} = 3.00 \text{ mm}.$$

Hence option **B**.

Question 19 — Correct option E**Topic: Optics.**

For the first single-slit minimum,

$$a \sin \theta = \lambda.$$

Therefore,

$$\sin \theta = \frac{500 \times 10^{-9}}{0.20 \times 10^{-3}} = 2.50 \times 10^{-3}.$$

Thus, $\theta \approx 2.50 \times 10^{-3} \text{ rad} = 0.143^\circ$. Option **E**.

Question 20 — Correct option C**Topic: Optics.**

Rayleigh's criterion for a circular aperture is

$$\theta_R = 1.22 \frac{\lambda}{D}.$$

Hence,

$$\theta_R = 1.22 \frac{550 \times 10^{-9}}{0.10} = 6.71 \times 10^{-6} \text{ rad}.$$

Therefore, option **C**.

3.2 Level II: Applied Problems**Question 21 — Correct option B****Topic: Oscillations.**

At $t = 0$,

$$x_0 = A \cos \phi, \quad v_0 = -A\omega \sin \phi.$$

Therefore,

$$A^2 = x_0^2 + \left(\frac{v_0}{\omega}\right)^2 = (0.030)^2 + (0.040)^2 = (0.050)^2,$$

so $A = 5.0 \text{ cm}$. Also,

$$\cos \phi = 0.030/0.050 = 0.6, \quad \sin \phi = -v_0/(A\omega) = 0.8.$$

Both are positive, so $\phi = 53.1^\circ$. Hence option **B**.

Question 22 — Correct option D**Topic: Oscillations.**

For a physical pendulum,

$$T = 2\pi \sqrt{\frac{I}{mgh}}.$$

For a uniform rod pivoted at one end, $I = mL^2/3$ and the centre of mass is at $h = L/2$. Thus,

$$T = 2\pi \sqrt{\frac{mL^2/3}{mg(L/2)}} = 2\pi \sqrt{\frac{2L}{3g}}.$$

With $L = 1.20 \text{ m}$,

$$T = 2\pi \sqrt{\frac{2(1.20)}{3(9.80)}} = 1.80 \text{ s}.$$

Therefore, option **D**.

Question 23 — Correct option C

Topic: Oscillations.

For three identical springs in series,

$$\frac{1}{k_{\text{eq}}} = \frac{1}{k} + \frac{1}{k} + \frac{1}{k} = \frac{3}{k},$$

so $k_{\text{eq}} = k/3 = 40 \text{ N m}^{-1}$. Therefore,

$$T = 2\pi\sqrt{\frac{m}{k_{\text{eq}}}} = 2\pi\sqrt{\frac{0.36}{40}} = 0.596 \text{ s}.$$

Hence option **C**.

Question 24 — Correct option A

Topic: Oscillations.

During the short collision, horizontal momentum is conserved:

$$m_b v_b = (M + m_b)V.$$

Thus,

$$V = \frac{(0.0100)(400)}{1.99 + 0.0100} = 2.00 \text{ m s}^{-1}.$$

After the collision, the total mass is 2.00 kg, so

$$\omega = \sqrt{\frac{k}{M + m_b}} = \sqrt{\frac{800}{2.00}} = 20.0 \text{ rad s}^{-1}.$$

Starting at equilibrium with speed V , the amplitude is $A = V/\omega = 2.00/20.0 = 0.100 \text{ m}$. Option **A**.

Question 25 — Correct option E

Topic: Oscillations.

Conservation of energy gives

$$\frac{1}{2}\kappa\theta_{\text{max}}^2 = \frac{1}{2}\kappa\theta^2 + \frac{1}{2}I\dot{\theta}^2.$$

Hence,

$$\dot{\theta} = \sqrt{\frac{\kappa}{I}(\theta_{\text{max}}^2 - \theta^2)}.$$

Substituting,

$$\dot{\theta} = \sqrt{\frac{0.80}{0.020} (0.25^2 - 0.15^2)} = \sqrt{40(0.0400)} = 1.26 \text{ rad s}^{-1}.$$

Thus, option **E**.

Question 26 — Correct option B

Topic: Normal Modes.

The symmetric-system frequencies are

$$\omega_1 = \sqrt{\frac{k}{m}}, \quad \omega_2 = \sqrt{\frac{k + 2k_c}{m}}.$$

Thus,

$$\omega_1 = \sqrt{\frac{100}{0.50}} = \sqrt{200} = 14.1 \text{ rad s}^{-1},$$

$$\omega_2 = \sqrt{\frac{100 + 100}{0.50}} = \sqrt{400} = 20.0 \text{ rad s}^{-1}.$$

Therefore, option **B**.

Question 27 — Correct option D**Topic: Normal Modes.**

From the in-phase frequency,

$$k = m\omega_1^2 = 2.0(10)^2 = 200 \text{ N m}^{-1}.$$

From the out-of-phase frequency,

$$k + 2k_c = m\omega_2^2 = 2.0(14)^2 = 392 \text{ N m}^{-1}.$$

Therefore,

$$2k_c = 392 - 200 = 192, \quad k_c = 96 \text{ N m}^{-1}.$$

Hence option **D**.**Question 28 — Correct option C****Topic: Normal Modes.**

Equating components gives

$$C_1 + C_2 = A, \quad C_1 - C_2 = 0.$$

The second equation gives $C_1 = C_2$. Substituting into the first gives $2C_1 = A$, so

$$C_1 = C_2 = \frac{A}{2}.$$

Therefore, option **C**.**Question 29 — Correct option A****Topic: Normal Modes.**

Direct substitution gives

$$q_1 = x_1 + x_2 = 3.0 + (-1.0) = 2.0 \text{ cm},$$

$$q_2 = x_1 - x_2 = 3.0 - (-1.0) = 4.0 \text{ cm}.$$

Thus, option **A**. The positive q_2 indicates a substantial out-of-phase component.**Question 30 — Correct option E****Topic: Normal Modes.**

The normal-mode condition is

$$\det(K - m\omega^2 I) = 0,$$

with

$$K - m\omega^2 I = \begin{bmatrix} k + k_c - m\omega^2 & -k_c \\ -k_c & k + k_c - m\omega^2 \end{bmatrix}.$$

Its determinant is

$$(k + k_c - m\omega^2)^2 - (-k_c)(-k_c) = (k + k_c - m\omega^2)^2 - k_c^2.$$

Hence option **E**.**Question 31 — Correct option B****Topic: Waves.**Here $k = 5 \text{ rad m}^{-1}$ and $\omega = 100 \text{ rad s}^{-1}$, so

$$v = \frac{\omega}{k} = 20 \text{ m s}^{-1}.$$

Since $v = \sqrt{T/\mu}$,

$$T = \mu v^2 = 0.010(20)^2 = 4.0 \text{ N}.$$

The average power is

$$P_{\text{av}} = \frac{1}{2} \mu \omega^2 A^2 v = \frac{1}{2} (0.010)(100)^2 (0.004)^2 (20) = 0.016 \text{ W}.$$

Thus, option **B**.

Question 32 — Correct option C**Topic: Waves.**

For a closed-open pipe,

$$f_m = \frac{(2m+1)v}{4L}, \quad m = 0, 1, 2, \dots$$

The fundamental is

$$f_1 = \frac{340}{4(0.85)} = 100 \text{ Hz.}$$

Only odd harmonics occur, so the next two are $3f_1 = 300 \text{ Hz}$ and $5f_1 = 500 \text{ Hz}$. Therefore, option **C**.

Question 33 — Correct option D**Topic: Waves.**

The increased tension raises the frequency to 505 Hz. For the same string length and linear density,

$$f \propto \sqrt{T}.$$

Therefore,

$$\frac{T_f}{T_i} = \left(\frac{f_f}{f_i}\right)^2 = \left(\frac{505}{500}\right)^2 = 1.0201.$$

The fractional increase is $1.0201 - 1 = 0.0201$. Hence option **D**.

Question 34 — Correct option A**Topic: Waves.**

For a moving source approaching a stationary observer,

$$f' = f \frac{v}{v - v_s}.$$

Thus,

$$f' = 600 \frac{343}{343 - 30} = 600 \frac{343}{313} = 658 \text{ Hz (approximately).}$$

The observed frequency is higher because the source approaches. Option **A**.

Question 35 — Correct option E**Topic: Waves.**

The Mach angle satisfies

$$\sin \mu = \frac{1}{M}.$$

For $M = 1.50$,

$$\mu = \sin^{-1} \left(\frac{1}{1.50} \right) = \sin^{-1}(0.6667) = 41.8^\circ.$$

Therefore, option **E**.

Question 36 — Correct option B**Topic: Optics.**

The sheet introduces additional optical path

$$\Delta L = (n - 1)t.$$

The number of fringe spacings shifted is

$$N = \frac{\Delta L}{\lambda} = \frac{(1.60 - 1)(5.0 \times 10^{-6})}{600 \times 10^{-9}} = \frac{3.0 \times 10^{-6}}{0.60 \times 10^{-6}} = 5.$$

Hence option **B**.

Question 37 — Correct option C**Topic: Optics.**

An interference maximum is missing when it coincides with a single-slit minimum. The condition is

$$m = p \frac{d}{a}.$$

Since $d/a = 4$,

$$m = 4p = 4, 8, 12, \dots$$

Therefore, option **C**.

Question 38 — Correct option D

Topic: Optics.

The line spacing is

$$d = \frac{1}{500 \text{ mm}^{-1}} = 0.00200 \text{ mm} = 2.00 \times 10^{-6} \text{ m}.$$

For $m = 2$,

$$\sin \theta_2 = \frac{2\lambda}{d} = \frac{2(600 \times 10^{-9})}{2.00 \times 10^{-6}} = 0.600,$$

so $\theta_2 = 36.9^\circ$. Also,

$$m_{\max} = \left\lfloor \frac{d}{\lambda} \right\rfloor = \lfloor 3.33 \rfloor = 3.$$

Hence option **D**.

Question 39 — Correct option A

Topic: Optics.

For critical incidence from glass to air,

$$\sin \theta_c = \frac{n_{\text{air}}}{n_{\text{glass}}} = \frac{1}{1.50},$$

so $\theta_c = 41.8^\circ$. For Brewster incidence from air to glass,

$$\tan \theta_B = \frac{n_{\text{glass}}}{n_{\text{air}}} = 1.50,$$

so $\theta_B = 56.3^\circ$. Therefore, option **A**.

Question 40 — Correct option E

Topic: Optics.

Bragg's law uses θ , not the displayed angle 2θ . Thus $\theta = 25.0^\circ$. For first order,

$$\lambda = 2d \sin \theta.$$

Therefore,

$$d = \frac{0.154}{2 \sin 25.0^\circ} \text{ nm} = 0.182 \text{ nm}.$$

Hence option **E**.

3.3 Level III: Multi-step Challenges

Question 41 — Correct option C

Topic: Oscillations.

Using $x = A \cos(\omega t + \phi)$ at $t = 0$,

$$\cos \phi = \frac{x_0}{A} = \frac{0.030}{0.060} = \frac{1}{2}.$$

Thus $\phi = \pm\pi/3$. The velocity is

$$v = -A\omega \sin(\omega t + \phi).$$

For $v(0) > 0$, we require $\sin \phi < 0$, so $\phi = -\pi/3$. Therefore,

$$x = 0.060 \cos(8t - \pi/3),$$

which is option **C**.

Question 42 — Correct option B

Topic: Oscillations.

The SHM speed-position relation is

$$v = \omega \sqrt{A^2 - x^2}.$$

Here,

$$\sqrt{A^2 - x^2} = \sqrt{(0.050)^2 - (0.030)^2} = \sqrt{0.0016} = 0.040 \text{ m}.$$

Thus,

$$\omega = \frac{0.40}{0.040} = 10 \text{ rad s}^{-1}, \quad T = \frac{2\pi}{\omega} = 0.628 \text{ s}.$$

Therefore, option **B**.

Question 43 — Correct option D

Topic: Oscillations.

The centre of mass is at $L/2$ from the end, while the pivot is at $L/4$, so $h = L/4$. The moment of inertia about the pivot is found by the parallel-axis theorem:

$$I = I_{\text{cm}} + mh^2 = \frac{mL^2}{12} + m\left\{\frac{L}{4}\right\}^2 = \frac{7mL^2}{48}.$$

Therefore,

$$T = 2\pi \sqrt{\frac{I}{mgh}} = 2\pi \sqrt{\frac{7L}{12g}} = 1.68 \text{ s}.$$

Hence option **D**.

Question 44 — Correct option A

Topic: Oscillations.

Write $x = A \cos(\omega t + \phi)$. At $t = 0$, $x = A/2$, so $\cos \phi = 1/2$. Moving toward equilibrium from the positive side means $v < 0$, hence $\sin \phi > 0$ and $\phi = \pi/3$. The opposite turning point occurs when the phase first reaches π :

$$\omega t + \frac{\pi}{3} = \pi \quad \Rightarrow \quad \omega t = \frac{2\pi}{3}.$$

Since $T = 2\pi/\omega$, $t = T/3$. Thus, option **A**.

Question 45 — Correct option E

Topic: Oscillations.

During sticking, horizontal momentum is conserved because the clay initially has no horizontal momentum:

$$Mv_0 = (M + m)V.$$

Thus,

$$V = \frac{(1.00)(2.00)}{1.25} = 1.60 \text{ m s}^{-1}.$$

The new angular frequency is

$$\omega' = \sqrt{\frac{k}{M + m}} = \sqrt{\frac{100}{1.25}} = 8.94 \text{ rad s}^{-1}.$$

At equilibrium all post-collision oscillator energy is kinetic, so $A' = V/\omega' = 1.60/8.94 = 0.179 \text{ m}$. Option **E**.

Question 46 — Correct option C

Topic: Normal Modes.

The frequencies are

$$\omega_1 = \sqrt{\frac{k}{m}} = 2, \quad \omega_2 = \sqrt{\frac{k + 2k_c}{m}} = \sqrt{16} = 4 \text{ rad s}^{-1}.$$

The initial displacement $[A, 0]^T$ contains equal modal coefficients:

$$x_1 = \frac{A}{2}(\cos \omega_1 t + \cos \omega_2 t), \quad x_2 = \frac{A}{2}(\cos \omega_1 t - \cos \omega_2 t).$$

At $t = \pi/2$, $\cos(2t) = \cos \pi = -1$ and $\cos(4t) = \cos 2\pi = 1$. Hence $x_1 = 0$ and $x_2 = -A$. Option **C**.

Question 47 — Correct option B

Topic: Normal Modes.

First convert to angular frequencies:

$$\omega_1 = 2\pi f_1 = 4\pi, \quad \omega_2 = 2\pi f_2 = 6\pi.$$

Then

$$k = m\omega_1^2 = 0.50(4\pi)^2 = 79.0 \text{ N m}^{-1}.$$

Also,

$$k + 2k_c = m\omega_2^2,$$

so

$$k_c = \frac{m(\omega_2^2 - \omega_1^2)}{2} = \frac{0.50[(6\pi)^2 - (4\pi)^2]}{2} = 49.3 \text{ N m}^{-1}.$$

Therefore, option **B**.

Question 48 — Correct option D

Topic: Normal Modes.

For the initial condition $[A, 0]^T$, the displacement of the first mass can be written

$$x_1 = A \cos\left(\frac{\omega_1 + \omega_2}{2}t\right) \cos\left(\frac{\omega_2 - \omega_1}{2}t\right).$$

Its amplitude envelope first becomes zero when

$$\cos\left(\frac{\Delta\omega}{2}t\right) = 0.$$

The first solution is $(\Delta\omega/2)t = \pi/2$, hence

$$t = \frac{\pi}{\Delta\omega} = \frac{\pi}{12 - 10} = 1.57 \text{ s}.$$

At this time the amplitude has transferred to the other mass. Option **D**.

Question 49 — Correct option A

Topic: Normal Modes.

Evaluate the energy at a turning point, where kinetic energy is zero. In phase, each outer spring stores $\frac{1}{2}kA^2$, while the coupling spring is unstrained:

$$E_{\text{in}} = kA^2.$$

Out of phase, the outer springs still store kA^2 in total, but the coupling spring changes length by $2A$:

$$E_{\text{out}} = kA^2 + \frac{1}{2}k_c(2A)^2 = (k + 2k_c)A^2.$$

Thus,

$$\frac{E_{\text{out}}}{E_{\text{in}}} = \frac{k + 2k_c}{k} = \frac{100 + 100}{100} = 2.$$

Hence option **A**.

Question 50 — Correct option E

Topic: Normal Modes.

Acting with K on the mode vectors gives

$$K \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad K \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 8 \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

For unit masses, $\omega_1 = \sqrt{4} = 2$ and $\omega_2 = \sqrt{8} = 2\sqrt{2}$. For the displacement decomposition,

$$C_1 + C_2 = 3, \quad C_1 - C_2 = 1,$$

so $C_1 = 2$ cm and $C_2 = 1$ cm. Thus, option **E**.

Question 51 — Correct option C

Topic: Waves.

Represent the three contributions as phasors:

$$A_R = A \left(1 + e^{i\pi/3} + e^{i2\pi/3} \right).$$

Using $e^{i\pi/3} = \frac{1}{2} + i\frac{\sqrt{3}}{2}$ and $e^{i2\pi/3} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$,

$$1 + e^{i\pi/3} + e^{i2\pi/3} = 1 + i\sqrt{3} = 2e^{i\pi/3}.$$

Therefore, the resultant amplitude is $2A = 10.0$ mm and its phase is $\pi/3$. Option **C**.

Question 52 — Correct option B

Topic: Waves.

The total time is the fall time plus the upward sound-travel time:

$$\sqrt{\frac{2h}{g}} + \frac{h}{v} = 4.00.$$

Let $u = \sqrt{h}$. Then

$$\frac{u^2}{343} + \sqrt{\frac{2}{9.80}}u - 4.00 = 0.$$

The positive root is $u = 8.399$, so

$$h = u^2 = 70.5 \text{ m}.$$

As a check, the fall time is 3.794 s and the sound time is 0.206 s, summing to 4.00 s. Hence option **B**.

Question 53 — Correct option E

Topic: Waves.

There are two Doppler shifts. First, the moving car is an observer approaching the stationary source:

$$f_1 = f_0 \frac{v + u}{v}.$$

The car then acts as a moving source approaching the detector:

$$f_2 = f_1 \frac{v}{v - u}.$$

Therefore,

$$f_2 = f_0 \frac{v + u}{v - u} = 40.0 \frac{343 + 30}{343 - 30} \text{ kHz} = 47.7 \text{ kHz}.$$

Thus, option **E**.

Question 54 — Correct option A

Topic: Waves.

The local amplitude is the magnitude of the position factor:

$$A_{\text{local}} = 0.080 \left| \sin \left(\frac{\pi(0.50)}{2} \right) \right| = 0.080 \sin \frac{\pi}{4} = 0.0566 \text{ m}.$$

At $t = 0.0625 = 1/16$ s,

$$\cos(4\pi t) = \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}.$$

Thus,

$$y = 0.080 \left(\frac{\sqrt{2}}{2} \right) \left(\frac{\sqrt{2}}{2} \right) = 0.0400 \text{ m.}$$

Hence option **A**.

Question 55 — Correct option D

Topic: Waves.

Far from the sources, the path difference is $\Delta L = d \cos \theta$ (an equivalent sine convention gives the same count). Maxima require

$$d \cos \theta = m\lambda.$$

Because $d/\lambda = 1.50/0.400 = 3.75$, the allowed integers are

$$m = -3, -2, -1, 0, 1, 2, 3,$$

seven values. For each value, $|m\lambda/d| < 1$, so $\cos \theta = m\lambda/d$ has two positions during a full circuit. Therefore, the total is $7 \times 2 = 14$. Option **D**.

Question 56 — Correct option B

Topic: Optics.

Substitute the specified values:

$$\left(\frac{\sin \beta}{\beta} \right)^2 = \left(\frac{1}{\pi/2} \right)^2 = \frac{4}{\pi^2},$$

and

$$\cos^2 \alpha = \cos^2 \frac{\pi}{3} = \left(\frac{1}{2} \right)^2 = \frac{1}{4}.$$

Therefore,

$$\frac{I}{I_0} = \frac{4}{\pi^2} \cdot \frac{1}{4} = \frac{1}{\pi^2} \approx 0.101.$$

Hence option **B**.

Question 57 — Correct option C

Topic: Optics.

The required resolving power is

$$R = \frac{\lambda}{\Delta\lambda} = \frac{600.0}{0.3} = 2000.$$

For a grating, $R = mN$. In second order,

$$N = \frac{R}{m} = \frac{2000}{2} = 1000.$$

At 500 lines per millimetre, the illuminated width is

$$W = \frac{1000}{500 \text{ mm}^{-1}} = 2.00 \text{ mm.}$$

Therefore, option **C**.

Question 58 — Correct option D

Topic: Optics.

The parameter is

$$\beta = \frac{\pi a \sin \theta}{\lambda}.$$

Thus,

$$\sin \theta = \frac{\beta \lambda}{\pi a} = \frac{4.493(600 \times 10^{-9})}{\pi(0.20 \times 10^{-3})} = 4.29 \times 10^{-3}.$$

Hence $\theta \approx 0.246^\circ$. The first secondary maximum has the standard intensity $I \approx 0.047 I_{\text{max}}$, or 4.7%. Therefore, option **D**.

Question 59 — Correct option A**Topic: Optics.**

Their small angular separation is

$$\theta \approx \frac{s}{L} = \frac{1.50}{100\,000} = 1.50 \times 10^{-5} \text{ rad.}$$

Rayleigh's criterion requires

$$\theta = 1.22 \frac{\lambda}{D}.$$

Therefore,

$$D = 1.22 \frac{550 \times 10^{-9}}{1.50 \times 10^{-5}} = 4.47 \times 10^{-2} \text{ m} = 4.47 \text{ cm.}$$

Hence option **A**.

Question 60 — Correct option E**Topic: Optics.**

The spacing is

$$d = \frac{1}{300 \text{ mm}^{-1}} = 3.333 \times 10^{-6} \text{ m.}$$

The largest possible positive order is

$$m_{\max} = \left\lfloor \frac{d}{\lambda} \right\rfloor = \left\lfloor \frac{3.333 \times 10^{-6}}{500 \times 10^{-9}} \right\rfloor = \lfloor 6.667 \rfloor = 6.$$

Because $a = d/3$, we have $d/a = 3$, so missing orders satisfy $m = 3p$: $m = 3$ and $m = 6$ within the allowed range. The observable positive orders are therefore 1, 2, 4, 5, four on each side. Including the central order gives $2(4) + 1 = 9$. Hence option **E**.

4 Compact Formula Checklist

Oscillations

$$F = -kx, \quad \omega = \sqrt{\frac{k}{m}}, \quad T = \frac{2\pi}{\omega},$$

$$x = A \cos(\omega t + \phi), \quad v = -A\omega \sin(\omega t + \phi),$$

$$a = -\omega^2 x, \quad E = \frac{1}{2}kA^2, \quad v^2 = \omega^2(A^2 - x^2).$$

$$T_{\text{simple pendulum}} = 2\pi\sqrt{\frac{L}{g}}, \quad T_{\text{physical}} = 2\pi\sqrt{\frac{I}{mgh}}.$$

Normal Modes

$$M\ddot{\mathbf{x}} + K\mathbf{x} = 0, \quad (K - \omega^2 M)\mathbf{A} = 0, \quad \det(K - \omega^2 M) = 0.$$

For the symmetric two-mass model,

$$\omega_1 = \sqrt{\frac{k}{m}}, \quad \mathbf{A}_1 \propto \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

$$\omega_2 = \sqrt{\frac{k + 2k_c}{m}}, \quad \mathbf{A}_2 \propto \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Waves and Sound

$$k = \frac{2\pi}{\lambda}, \quad \omega = 2\pi f, \quad v = f\lambda = \frac{\omega}{k}.$$

$$v_{\text{string}} = \sqrt{\frac{T}{\mu}}, \quad P_{\text{av}} = \frac{1}{2}\mu\omega^2 A^2 v.$$

$$f_n = \frac{nv}{2L} \quad (\text{fixed-fixed/open-open}),$$

$$f_m = \frac{(2m+1)v}{4L} \quad (\text{closed-open}).$$

$$f' = f \left(\frac{v \pm v_o}{v \mp v_s} \right), \quad \sin \mu = \frac{1}{M}.$$

Optics

$$n_1 \sin \theta_1 = n_2 \sin \theta_2, \quad v = \frac{c}{n}, \quad \lambda = \frac{\lambda_0}{n}.$$

$$\Delta y_{\text{double slit}} = \frac{\lambda L}{d}, \quad \text{single-slit minima: } a \sin \theta = m\lambda.$$

$$\theta_R = 1.22 \frac{\lambda}{D}, \quad \text{grating maxima: } d \sin \theta = m\lambda.$$

$$R = mN, \quad n\lambda = 2d \sin \theta \quad (\text{Bragg}).$$